

Code generation for memory monitoring

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1 Introduction

Memory accesses from a pointer or an array are not safe in C, thus trying to access an element of an array out of valid range or an invalid pointer may cause a segmentation fault during execution.

C compilers such as GCC do not provide detection mechanisms for this kind of errors. We propose an automatic code instrumentation, so that the generated code will perform memory monitoring and will be able to check memory properties during execution (runtime checking). We consider as readable and writable memory: global variables, dynamically allocated memory, and formal parameters and local variables of each function. We decided to use the term “block” to nominate these four cases.

E-ACSL is an executable subset of ACSL [1], a formal specification language for C using source code annotations to express properties. ACSL-annotated C programs can be dealt with FRAMA-C [2], a framework for modular analysis of C. FRAMA-C provides a plug-in generating executable C code from E-ACSL annotations, used by the implementation discussed in this paper.

2 Related work

3 Stored information

Memory blocks (as defined above) held properties such as their base address and their size. E-ACSL provides five annotations to retrieve useful information about blocks:

`\base_addr(p)` returns the base address of the block containing pointer p

`\block_length(p)` returns the size (in bytes) of the block containing p

`\offset(p)` returns the offset between p and `\base_addr(p)`

`\valid_read(s)` whether reading $*s$ is safe

`\valid(s)` whether reading and writing `*s` is safe

`\initialized(s)` whether `s` has been initialized

Our contribution allowed the E-ACSL plug-in to generate executable C code from these five annotations, thus checking these five memory properties during execution.

For each block, we need to store the following information: the base address (address of the first element within the block), the size (in bytes), the validity status (whether reading or writing the block is safe) and the initialization status for each byte of the block. We call *block descriptor* the data structure storing this information.

To keep the memory space occupied by this structure low, whenever none or all of the bytes of a memory block have been initialized (the two most common cases), the initialization array is freed; an integer field counting initialized bytes is used instead.

4 Global architecture

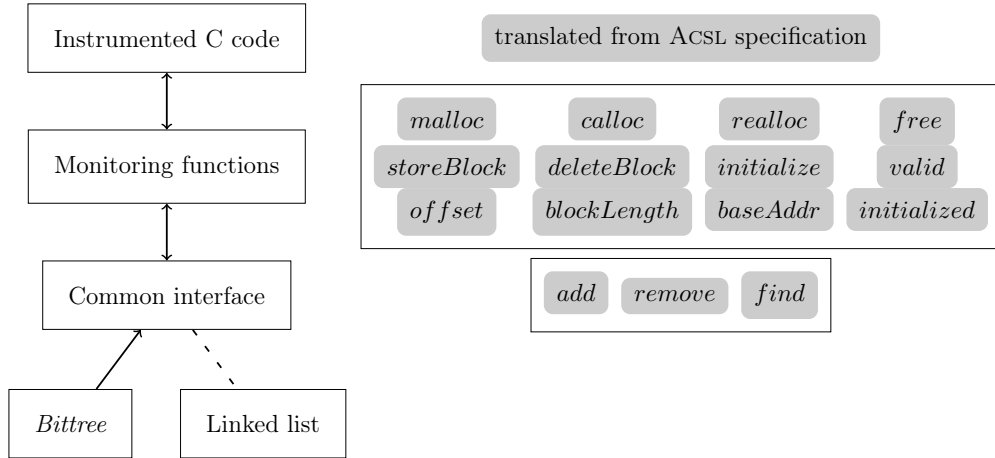


Figure 1: Global architecture

Interactions between the components of our solution are displayed by Figure 1. Informations about block descriptors are stored in an arbitrary data structure S (see Section 5). The E-ACSL plug-in performs an instrumentation to generate executable C code invoking monitoring functions (see Section 6) relying on the data structure S .

5 Concrete data structure used: *bittree*

In this section we discuss the choice of the concrete data structure used to store the block descriptors.

5.1 *Patricia tries*

We need a data structure with a good time and space complexity, indeed we may have to often add or remove a block descriptor in the structure. The structure has to be sorted: we want to access to a block descriptor by its base address, and also to its predecessor and successor. Thus, hash-tables will not fit. We also unconsidered the linked lists, due to the linear worst-case complexity. The (unbalanced) binary search trees provide a linear worst-case complexity too when the base address of inserted elements are monotonically increasing, and this may be quite common. Self-balanced binary search trees are dismissed because of the numerous add/remove operations implying numerous costly balancing operations.

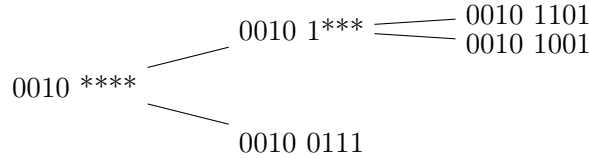


Figure 2: Example of bittree

We choose to use the *Patricia tries* [3] structure, which is efficient even if the tree is unbalanced. The prefix used by a node (not a leaf) is the greatest common prefix (on 32 or 64 bits) of its two children. The block descriptors are held on leaves. Other nodes just do the routing from the root to a block descriptor. Patricia tries are usually used on strings and characters, so we named “bittree” the structure used in this article. For example on 8-bit addresses, Figure 2 shows a bittree storing three block descriptors (identified by their addresses: 0010 0111, 0010 1001 and 0010 1101). The greatest common prefix of 0010 1001 and 0010 1101 is 0010 1 *** and the greatest common prefix of 0010 0111 and 0010 1 *** is 0010 ****. The * means that this bit is meaningless. Figure 3 shows that a new node (with a new prefix) is added when a block descriptor is inserted.

Figure 4 shows an example of deletion of a 8-bit block descriptor. Patricia tries being compact prefix trees, a node having an only child is deleted. So 0010 1 *** becomes useless and is replaced by its child 0010 1001. Deleting useless nodes, or only storing useful ones for routing, keeps the tree exploration efficient. A 32-bit (respectively 64-bit) bittree has a worst-case depth of 33 (respectively 65) nodes.

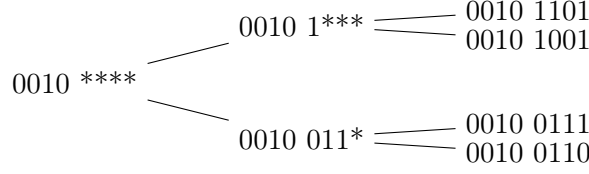


Figure 3: Bittree after adding 0010 0110 into the bittree of Fig. 2

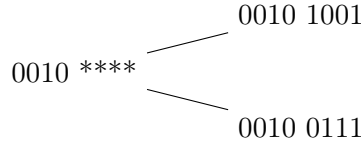


Figure 4: Bittree after deleting 0010 1101 from the bittree of Fig. 2

5.2 Greatest common prefix computation

The greatest common prefix of A and B can be naively computed by: $X = \neg(A \oplus B)$ (where $\oplus$ is the XOR operator) to keep bits in common, then each bit of X on the right side of a 0 is set to 0. The obtained mask is then applied to A or B . For example, considering $A = 0110\,0111$ and $B = 0111\,1111$, $X = \neg(A \oplus B) = 1110\,0111$, setting each bit on the right side of a 0 to 0 we get $1110\,0000$. We apply this mask to A and get the greatest common prefix of A and B : $011\,*\,*\,*\,*$. This algorithm is used by the first version of our implementation and is named *Bittree-naive* in the experiments.

We optimized this algorithm by firstly computing all of the 65 or 33 different masks (from 0×0 to $0 \times f \dots f$) and storing them in an array. Then we use a dichotomic search to find the mask corresponding to the greatest common prefix: if A and B have a 32-bit common prefix, do they have a 48-bit common prefix? Otherwise, do they have a 16-bit common prefix? And so on. This search takes at most 6 (respectively 5) steps on 64-bit (respectively 32-bit) bittrees. This algorithm is used by the final version of our implementation and is named *Bittree-opti* in the experiments.

5.3 Experiments

These experiments justify the choice of bittrees over linked lists and binary search trees. We implemented the classic *merge sort* algorithm, and added extra allocations/deallocations to put each data structure to the test. The program has been instrumented to call our monitoring functions (see Section 6). The execution time of the instrumented program using each data structure has been measured (in micro-seconds) and is plotted against the number of calls to a function *store* (adding an element to the data structure).

Figure 5 displays the results of the experiments. The reference time is the execution time of the program without any instrumentation. Bittree-naive and Bittree-opti are using two different versions of the greatest common prefix computation (see Sub-section 5.2).

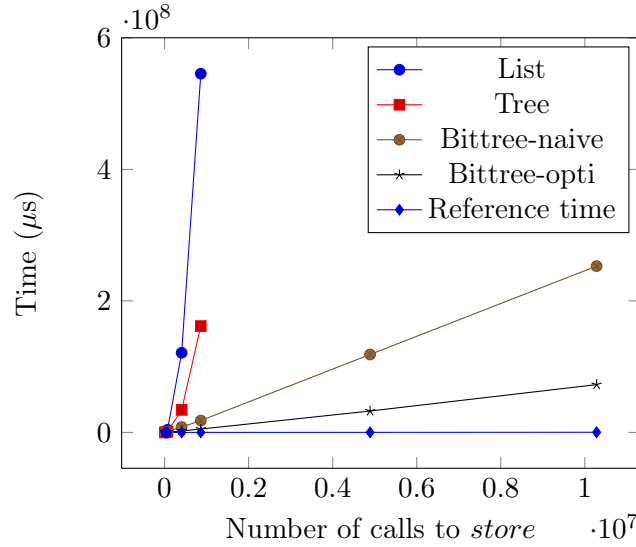


Figure 5: Execution time plotted against the number of calls to *store*

The last but one measurement, the merge sort applied to 50,000 elements (4,889,819 calls to *store*), has a reference time of 0.07s. It runs 32s with the optimized bittree, 118s with the naive bittree, 7 hours 15 minutes with the binary search trees and 19 hours 15 minutes with linked lists. Our last measurement, the merge sort applied to 100,000 elements (10,279,851 calls to *store*) has a reference time of 0.19s. It runs 72s with the optimized bittree, 252s with the naive bittree, but binary search trees and linked lists exceed our 24-hour timeout.

6 Monitoring functions

This section presents the functions used for adding/removing/retrieving in formations about block descriptors into our data structure (bittree). These functions will be automatically inserted in the source code by the instrumentation performed by the E-ACSL plug-in.

6.1 Automatic allocation

Automatic allocation functions keep track of all non-dynamically allocated variables (but occupying memory space nevertheless), such as formal param-

eters, local variables or global variables.

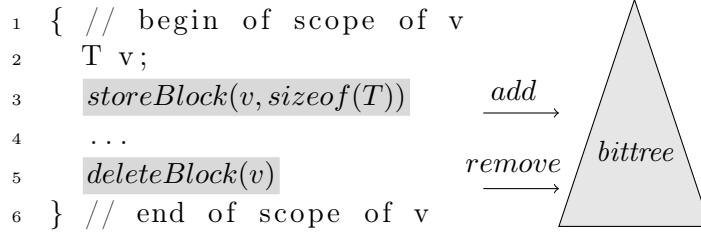


Figure 6: store and delete block

6.2 Dynamic allocation

These functions have to be used instead of *malloc*, *realloc*, *calloc* and *free* from the standard library.

6.3 Initialization

These functions have to be used for each assignment, to update the initialization status of a block descriptor. *initialize(ptr, size)* marks the *size* bytes starting from *ptr* as initialized. *fullInit(ptr)* marks all the bytes of *ptr* as initialized at once. It is designed to avoid multiple calls to *initialize* whenever it is possible and improves efficiency of the instrumented program.

6.4 Interrogation

These functions are used to retrieve information about the block descriptors and match the E-ACSL annotations we are trying to support: `\valid`, `\valid_read`, `\base_addr`, `\block_length`, `\offset` and `\initialized`.

valid(ptr, size) returns 1 if it is safe to read/write *size* bytes starting from *ptr*, 0 otherwise. *baseAddr(ptr)* returns the base address of the block containing *ptr* if such a block exists, NULL otherwise. *blockLength(ptr)* returns the size (in bytes) of the block containing *ptr* if such a block exists, 0 otherwise. *offset(ptr)* returns the offset between *ptr* and the base address of the block containing *ptr* if such a block exists, -1 otherwise. *initialized(ptr, size)* returns 1 if the *size* first bytes starting from *ptr* are initialized, 0 otherwise.

7 Experimental results

	alarms	τ_{exec}	mutants	mutation	test cases	assert violations
p_1	19	63s	6	pointer arith.	57	156/186 (83.8 %)
p_2	8	28s	2	pointer arith.	27	52/54 (96.3 %)

Figure 7: Experimental results

8 Conclusion

We implemented an efficient data structure (bittree) to store the block descriptors and functions relying on this structure to perform memory monitoring. We also defined and implemented into the E-ACSL plug-in the instrumentation to perform on a C source code to monitor the memory. This allows the runtime checking of the following E-ACSL annotations: `\base_addr`, `\block_length`, `\offset`, `\valid`, `\valid_read` and `\initialized`.

References

- [1] P. Baudin, P. Cuoq, J. C. Filliâtre, C. Marché, B. Monate, Y. Moy, and V. Prevosto. ACSL: ANSI/ISO C Specification Language preliminary design version 1.5. <http://frama-c.com/acsl.html>, 2010.
- [2] P. Cuoq, F. Kirchner, N. Kosmatov, V. Prevosto, J. Signoles, and B. Yakobowski. Frama-C: A program analysis perspective. In *Proc. of the 10th International Conference on Software Engineering and Formal Methods*, SEFM ’2012, October 2012. To appear.
- [3] W. Szpankowski. Patricia tries again revisited. *J. ACM*, 37(4):691–711, Oct. 1990.